

bioETH-PRS: Confidential Polygenic Risk Scoring with Smart Contracts on an FHE-Enabled Blockchain

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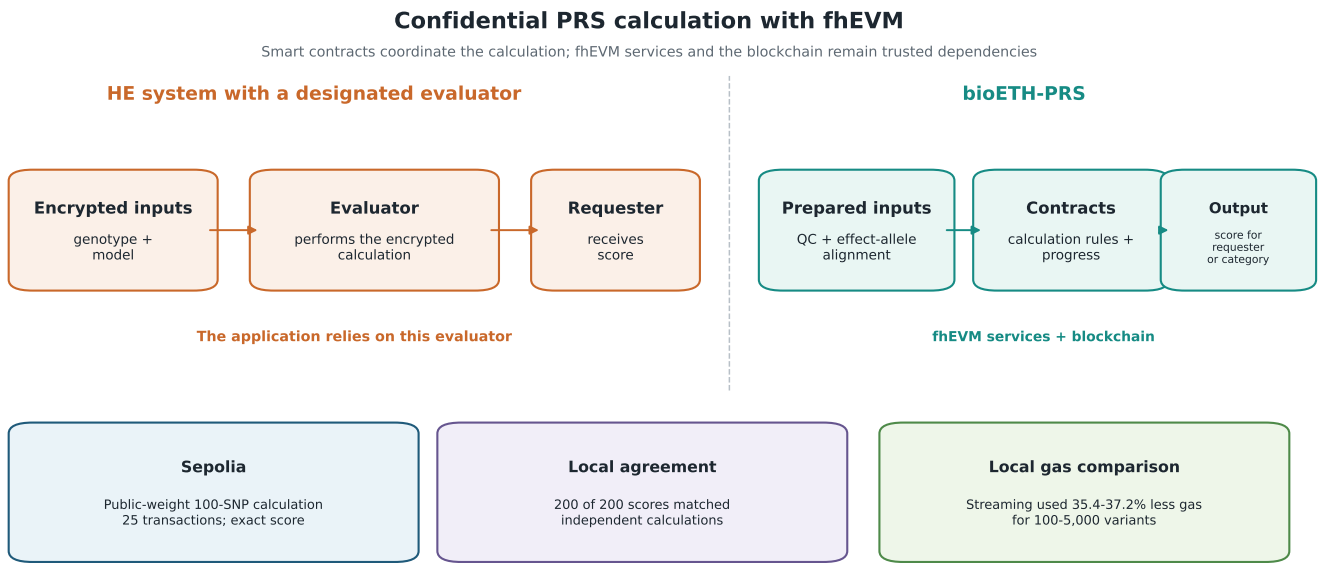


Figure 1: Graphical Abstract. bioETH-PRS uses smart contracts to coordinate an encrypted PRS calculation instead of relying on a designated evaluator. The system still depends on the blockchain and on fhEVM services that perform encrypted calculations and decrypt results for the intended requester. We report a public-weight 100-SNP calculation on Sepolia, exact agreement for 200 calculations in a local simulation, and 35.4–37.2% lower gas for the Streaming method in the local simulation.

Abstract

Polygenic risk scores (PRSs) aggregate genetic effect estimates to predict disease susceptibility. Calculating a PRS with a conventional external service can require sharing raw genotype data. HEPRS, the homomorphic-encryption system used for comparison in this study, requires a designated evaluator. We present **bioETH-PRS**, which uses publicly auditable smart contracts to coordinate the calculation on a blockchain that supports Fully Homomorphic Encryption (fhEVM). This reduces reliance on a single evaluator, but the system still depends on the smart contracts, blockchain, and fhEVM computation and decryption services. Before encryption, the input is checked for valid genotype values and aligned to the model’s effect allele. bioETH-PRS converts signed weights to nonnegative integers without overflow. The model provider can require a randomized risk category instead of the raw score. One public-weight 100-SNP calculation completed on Sepolia in 25 transactions and used 20.710271 million gas. It took 269,320 ms from submission to result and 8,081 ms to decrypt; the encoded score of 758,685 matched an independent calculation exactly. One private-weight 100-SNP calculation was evaluated only in a local simulation. Across 200 local public-weight calculations with 100–5,000 variants, every decoded score matched the independent calculation because the study weights were represented without rounding. The Streaming method processes each group without permanently storing its inputs; it used 35.4–37.2% less local gas than the Classic method (stored inputs). This study evaluates additive PRS models containing up to 5,000 variants; we did not evaluate genome-wide or clinical use.

Key Points

- Smart contracts reduce reliance on a single designated evaluator, but the system still depends on the blockchain and fhEVM services.
- A public-weight 100-SNP calculation completed on Sepolia. In the local simulation, we evaluated public-weight calculations with 100–5,000 variants and one private-weight 100-SNP calculation.
- All 200 locally decoded public-weight scores matched Equation 1 exactly because the study weights were represented without rounding.
- The Streaming method used 35.4–37.2% less gas than the Classic method (stored inputs) in the local simulation.
- Randomized risk categories and query limits make exact recovery of private weights harder, but they do not provide differential privacy or completely hide the weights.

1 Introduction

Advances in genome sequencing have made polygenic risk scores (PRSs) a central tool in precision medicine [Wray et al., 2021, Inouye et al., 2018]. A PRS aggregates the dosage-weighted effects of genetic variants:

$$\text{PRS} = \sum_{i=1}^N g_i \cdot \beta_i, \quad (1)$$

where $g_i \in \{0, 1, 2\}$ is the dosage of the model-specified effect allele for variant i and β_i is the corresponding GWAS effect weight. Published polygenic scores in the PGS Catalog range from small panels to genome-wide scores with large variant sets [Lambert et al., 2019]. Using a conventional external service to calculate a PRS can require sending genotype data outside the hospital, which raises privacy concerns because genomic sequences can be identifying [Gymrek et al., 2013, Erlich and Narayanan, 2014].

Two kinds of sensitive information. Two kinds of sensitive information require simultaneous protection: (1) *patient genotypes* (g_i), identifying, heritable, and permanent; and (2) *GWAS model weights* (β_i), derived from restricted research cohorts. More generally, model interfaces and outputs can leak sensitive information through model-inversion attacks [Fredrikson et al., 2015]. In a conventional plaintext service, the evaluator must decrypt both genotypes and weights before the calculation can proceed.

Why FHE is useful. Fully Homomorphic Encryption (FHE) permits arbitrary arithmetic on encrypted data without decryption. Knight et al. [Knight et al., 2026] demonstrate this with HEPRS, an open-source system that computes a 110,000-SNP schizophrenia PRS under the CKKS scheme [Cheon et al., 2017] with near-zero accuracy loss ($r > 0.999$, $\text{MSE} < 2.3 \times 10^{-6}$). HEPRS, however, still uses a central evaluator: it relies on a three-party trust model (client, modeler, evaluator), and a colluding evaluator and client can reveal GWAS model weights. Trust in the evaluator is an assumption, not a protocol guarantee. Furthermore, HEPRS requires 3–4 GB of RAM per individual even at 110,000 SNPs, and output is a raw score—there is no built-in mechanism to prevent score accumulation or model probing.

Our contribution. We propose **bioETH-PRS**, in

which publicly auditable smart contracts coordinate the encrypted calculation and control how results are released. A designated evaluator is no longer required. However, the system still relies on the contracts, blockchain consensus, and the fhEVM computation and decryption services (Figure 1). The blockchain records the contract steps, but it does not by itself prove that the encrypted arithmetic was correct.

Distinctions from HEPRS. Although both systems calculate the same encrypted PRS, they use different methods. First, bioETH-PRS converts signed decimal weights to unsigned integers for TFHE/fhEVM; HEPRS uses CKKS approximate decimal arithmetic. Second, four public contracts manage who may use a sample, model publication, calculation, and result release. Third, the Classic method (stored inputs) keeps the encrypted SNPs for later calculation, whereas the Streaming method processes each group immediately. Fourth, a model can require a randomized risk category and limit repeated queries instead of releasing the raw score.

Our main contributions are:

- Four smart contracts that separately manage who may use a sample, model publication, encrypted calculation, and result release.
- A three-step conversion that represents signed decimal weights as nonnegative integers, checks that the calculation stays within the allowed integer range, and selects a suitable scale before the model is published.
- The Classic method (stored inputs) and Streaming method for processing the encrypted inputs; the Streaming method used about 37% less gas in the local simulation.
- A randomized risk category that can be required instead of releasing the raw score.
- Query limits and thresholds fixed by the model provider, evaluated with queries chosen after earlier results or in advance, several wallets and samples, and correlated SNP patterns.
- A public-weight 100-SNP calculation on Sepolia, public-weight calculations with 100–5,000 variants in the local simulation, and one local private-weight 100-SNP calculation.

The intended use is research with additive PRS models containing up to 5,000 variants. We did not evaluate genome-wide or clinical use, production cost, or private weights on Sepolia.

2 Background

2.1 Polygenic Risk Scores

A PRS is the weighted inner product of a patient’s effect-allele dosage vector $\mathbf{g} \in \{0, 1, 2\}^N$ and a GWAS effect weight vector $\boldsymbol{\beta} \in \mathbb{R}^N$ (Eq. 1). Effect weights are typically estimated by large-scale GWAS over tens of thousands of individuals and are signed floating-point values in the range $[-0.5, +0.5]$. Standard tools such as PLINK 2 [Chang et al., 2015] and LDpred2 [Privé et al., 2021] compute PRSs in milliseconds on plaintext data; FHE approaches trade runtime for privacy.

PRS models span a wide range in SNP inclusion depending on construction methodology. Sparse models may include hundreds to a few thousand variants, while genome-wide approaches incorporate tens of thousands to millions. We evaluated bioETH-PRS with models containing up to 5,000 variants. These results do not establish genome-wide or clinical use.

2.2 Genotype preprocessing, QC, and model alignment

bioETH-PRS scores one individual against an already developed model. Minor-allele frequency and Hardy–Weinberg filtering are therefore cohort- and model-development quality-control operations performed upstream when effect weights are derived. The scoring procedure instead checks missingness, genome build, variant identity and order, allele orientation, and the dosage representation before any value is encrypted.

Only diploid hard-call dosages in $\{0, 1, 2\}$ are accepted. A non-integer or out-of-range dosage is rejected rather than clamped, because clamping silently changes the score. The model must state one of three rules for missing values: reject the sample, use dosage 0, or use a supplied cohort mean rounded half away from zero to a hard call in $\{0, 1, 2\}$. An unstated zero is avoided because it would be indistinguishable from a genuine homozygous-reference call. The declared genotype build must equal the model build; a mismatch is fatal because the build cannot be inferred from dosage values. Variant identifiers and order are checked element by element, not merely by vector length. Duplicate identifiers, multiallelic sites, and indels are rejected; this study evaluates biallelic SNP hard calls only.

Effect-allele alignment does not reveal encrypted weight magnitudes. Publicly visible model information supplies each variant identifier, genome build, effect allele, other allele, and column order, and alignment runs locally before encryption. For a biallelic, non-palindromic SNP, a dosage already counting the effect allele is retained; a

dosage counting the other allele is transformed as $g_{\text{effect}} = 2 - g$. A compatible strand complement is applied before the same decision. Palindromic A/T or C/G pairs without explicit strand resolution are rejected even on a literal allele match, because the same label is consistent with opposite strand orientations. Any remaining incompatible pair is rejected.

The preparation record reports how many variants were matched, missing, imputed, rejected, or changed for allele orientation. The HEPRS data sets contain numeric matrices without variant identifiers, builds, or allele labels and are therefore treated as already aligned. Their leading constant column has weight 0 and dosage 1, so an N -variant data set contains $N + 1$ encoded positions. Examples with variant metadata covered the build, order, missingness, and strand rules.

2.3 Fully Homomorphic Encryption

FHE schemes differ in the arithmetic they support and their performance. Two are directly relevant to this work.

CKKS [Cheon et al., 2017] supports approximate arithmetic over real-valued data, is naturally suited to decimal calculations, and underpins HEPRS. Its available calculation depth depends on the selected parameters and whether bootstrapping is used, and approximation error accumulates during calculation. The Lattigo library [Tune Insight, 2023] implements CKKS in Go with the parameter sets used by HEPRS. CKKS is computationally expensive: evaluating a 110,000-SNP PRS on a single CPU node requires approximately 4.9 s per individual and up to 130 GB of RAM for a cohort of 1,000 [Knight et al., 2026].

TFHE [Chillotti et al., 2020] supports bootstrapped operations on encrypted values. The fhEVM interface used here exposes exact unsigned-integer operations to smart contracts. Signed decimal weights must therefore be converted to unsigned integers, motivating Section 4.

The security of TFHE is based on LWE/GLWE-family assumptions [Chillotti et al., 2020]; CKKS uses related lattice assumptions. These assumptions are generally studied as candidates for security against quantum attacks, but the present study does not evaluate long-term cryptographic parameter selection.

2.4 Programmable Blockchain and fhEVM

The Fully Homomorphic Ethereum Virtual Machine (fhEVM) extends the Ethereum execution environment with TFHE precompile operations [Zama, 2025]. Every encrypted value is represented on-chain by a 32-byte opaque handle; actual ciphertexts are managed by an off-chain coprocessor. Smart contracts store only handles and trigger FHE operations through coprocessor calls.

An on-chain contract records who may decrypt each result. If the fhEVM services work as specified, only the intended requester can decrypt it. The blockchain records these decisions and calls, but it does not independently

verify the encrypted calculation or prevent a failure of the fhEVM calculation and decryption services.

A per-transaction Homomorphic Computation Unit (HCU) budget limits FHE operations. In the local simulation, the largest successful calculation chunk contained 21 SNPs for both model types (Section 7); the Sepolia limit remains unmeasured and is not inferred from that result. The HCU constraint necessitates chunked computation strategies for large input vectors.

3 System design

3.1 System overview

bioETH-PRS uses four smart contracts to move from sample registration to a risk result (Figure 2). The registry records who may use a sample, the Model contract stores the GWAS model and its result settings, the Calculation contract calculates the encrypted score, and the Result contract can release a randomized risk category. Separating these roles makes the responsibility of each contract clear.

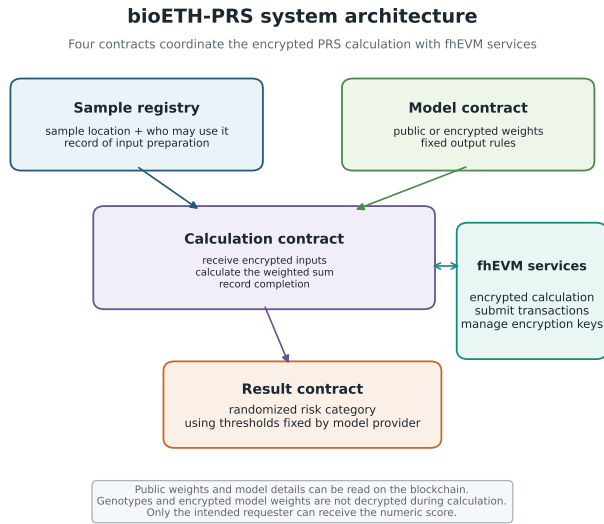


Figure 2: System architecture. Four contracts record who may use a sample, store the model and its result settings, calculate the encrypted score, and return either the score or a randomized risk category. Public weights and model details are visible by design; genotype dosages and private weights remain encrypted during the calculation. Results are released only to the intended requester.

Sample registry. The registry stores references (IPFS or Arweave URIs) to encrypted genotype data held off chain. A requester registers a sample and chooses who may use it in a calculation. The registry keeps this record but performs no encrypted arithmetic.

Model contract. The model contract stores GWAS weights in groups. Public weights are visible integers; private weights are encrypted. Public weights are encrypted before multiplication in the present implementation. The

model record also stores the scale factor, weight shift, score shift, and a cryptographic fingerprint of the input-preparation record. Models cannot be changed after publication. Public-weight calculations with 100–5,000 variants and one private-weight 100-SNP calculation were evaluated locally. Only the public-weight 100-SNP calculation was performed on Sepolia.

Calculation contract. The calculation contract receives the encrypted SNPs in groups, calculates the weighted sum, corrects the integer shifts described in Section 4, and releases one result. The final encrypted score can be decrypted only by the intended requester. Section 5 describes the two calculation methods.

Result contract. The result contract adds a random value to the encrypted score before assigning Low, Medium, or High risk. The random value comes from the fhEVM and is not known before the transaction is mined. The model provider can require this category result instead of allowing the requester to decrypt the raw score.

3.2 Comparison with HEPRS

Table 1 compares bioETH-PRS and HEPRS [Knight et al., 2026] by how they handle encrypted data, model size, runtime, memory, deployment needs, and the result returned. Both compute Equation 1. HEPRS reports encrypted CKKS results for a much larger model with a designated evaluator. bioETH-PRS reports a smaller calculation coordinated by public smart contracts. These contracts reduce reliance on one evaluator, but the system still depends on the blockchain and fhEVM services.

4 Representing decimal weights as integers

4.1 Why conversion is needed

This implementation represents weights and scores as unsigned 64-bit integers. GWAS weights β_i are signed floating-point values; a naive fixed-point encoding that multiplies by a scale factor s and rounds produces quantized weights $q_i = \text{round}(s \cdot \beta_i) \in \mathbb{Z}$, which may be negative. Negative integers cannot be stored in an unsigned 64-bit integer and would silently wrap under modular overflow, corrupting results without any observable error. A correct encoding must guarantee that all intermediate encrypted values remain in the non-negative range $[0, 2^{64} - 1]$ for any genomically valid input $g_i \in \{0, 1, 2\}$.

4.2 Three-step conversion

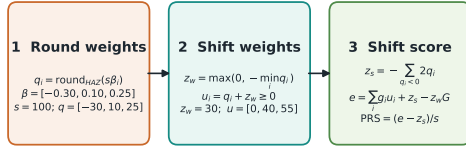
We convert the signed decimal weights to nonnegative integers in three steps (Figure 3). The integer shifts are reversible. Conversion of the original decimal weights is ex-

Table 1: Comparison with HEPRS. HEPRS values come from the published study; bioETH-PRS values state whether they were observed on Sepolia, observed in the local simulation, or not measured.

Dimension	HEPRS [Knight et al., 2026]	bioETH-PRS
How encrypted data are handled	encrypted client-modeler-evaluator system	encrypted inputs; contracts determine who receives the result
Designated evaluator	yes; evaluator executes CKKS	no designated evaluator; contracts coordinate the calculation
Remaining dependencies	three-party non-collusion and HE software/hardware	contracts, blockchain, fhEVM computation, and decryption services
Arithmetic scheme	CKKS approximate signed-real arithmetic	TFHE unsigned integers after fixed-point encoding
Encrypted variants evaluated	110,000 in the published study	public: 100 on Sepolia and 100–5,000 locally; private: 100 locally
Runtime	approximately 4.9 s/person at 110,000	269.320 s to result plus 8.081 s to decrypt at 100 on Sepolia; local times are not comparable
Memory	3–4 GB/person; up to 130 GB for 1,000	not measured
Deployment needs	evaluator compute host and client/modeler setup	fhEVM blockchain, computation service, relay, and decryption service
Result returned	raw score delivered to client; external controls optional	raw score or randomized risk category, as set by the model provider
Public information	model information managed off chain	model identifiers, alleles, order, result settings, and transactions are public; weights may be encrypted

Converting signed decimal weights to nonnegative integers

Weights are rounded and shifted; decoding restores the score after calculation



Worked example begins with Equation 1
 $g = [0, 2, 1]$; PRS from Equation 1 = 0.45
 shifted weighted sum = 135; $G = 3$; $z_s = 60$
 encoded $e = 105$; decoded PRS = 0.45

Safety condition for all values used in the calculation: $4sMN \leq 2^{64} - 1$

Figure 3: Three-step conversion of signed weights to non-negative integers. Half-away-from-zero rounding and $z_w = \max(0, -\min_i q_i)$ are shown explicitly. The worked example is converted back to the original additive PRS.

act only when the selected scale represents those weights without rounding, as it does for the data sets used here.

Step 1: Scale.

$$q_i = \text{round}_{\text{away}}(s \cdot \beta_i), \quad q_i \in \mathbb{Z}, \quad (2)$$

where s is a model-specific scale factor selected by the quantization procedure (Section 4.5) and exact half ties are rounded away from zero. The study data sets require no tie breaking: all weights have at most six decimal places and the selected scales are integer multiples of 10^6 , so their quantization is exact.

Step 2: Weight zero-point shift.

$$z_w = \max(0, -\min_i q_i), \quad u_i = q_i + z_w \geq 0 \quad \forall i. \quad (3)$$

The clamp is required by the unsigned on-chain represen-

tation when all weights are positive; algebraically, $z_w = 0$ already preserves $u_i \geq 0$. The shifted weights $u_i \geq 0$ are stored on-chain. The raw accumulation $\sum_i g_i u_i$ overestimates the true PRS by the constant $z_w \cdot \sum_i g_i$, requiring correction. The engine tracks the genotype sum $G = \sum_i g_i$ as a second encrypted accumulator to enable this correction.

Step 3: Score zero-point shift. Even after the weight correction, the corrected sum $P = \sum_i g_i u_i - z_w G$ can be negative for patients with many risk-decreasing alleles. We define:

$$z_s = -\sum_{i: \beta_i < 0} 2q_i = \sum_{i: \beta_i < 0} 2|q_i|, \quad (4)$$

$$e = P + z_s \geq 0. \quad (5)$$

e is the *encoded score* stored encrypted on-chain. The on-chain computation is structured to avoid any intermediate negative value:

$$e = (\text{partialSum} + z_s) - (z_w \cdot G), \quad (6)$$

where z_s is added before subtracting $z_w G$, ensuring the intermediate value $(\text{partialSum} + z_s)$ is always non-negative.

Decoding. After decryption by the intended requester:

$$\text{PRS} = \frac{e - z_s}{s}. \quad (7)$$

4.3 Worked Example

Consider three genetic variants with scale $s = 100$. Their weights and effect-allele dosages are

$$\beta = [-0.30, 0.10, 0.25], \quad \mathbf{g} = [0, 2, 1].$$

The plaintext target is

$$\text{PRS} = 0(-0.30) + 2(0.10) + 1(0.25) = 0.45.$$

Six steps in PRS calculation

Variant identifiers are aligned before genotypes and model weights are encrypted

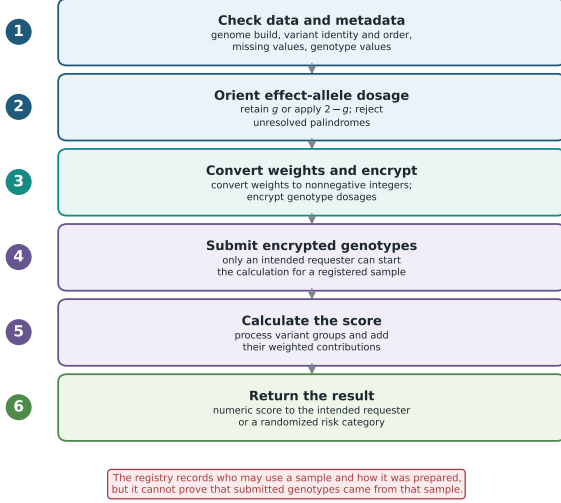


Figure 4: Six steps in the PRS calculation. Public identifiers and allele labels support local QC and effect-allele orientation before dosage values and, for private weights, weight magnitudes are encrypted. The contracts record who receives the result, but they do not prove that the encrypted values came from the registered sample.

1. Quantize: $\mathbf{q} = [-30, 10, 25]$.
2. Weight shift: $z_w = 30$, $\mathbf{u} = [0, 40, 55]$.
3. Accumulate: $\text{partialSum} = 0 \cdot 0 + 2 \cdot 40 + 1 \cdot 55 = 135$;
 $G = 0 + 2 + 1 = 3$.
4. Correction: $P = 135 - 30 \cdot 3 = 45$.
5. Score shift: $z_s = 2 \cdot 30 = 60$; $e = 45 + 60 = 105$.
6. Decode: $\text{PRS} = (105 - 60)/100 = 0.45$.

4.4 Checking the integer range

Proposition 1. Let N be an arbitrary SNP count, let s be the fixed-point scale factor, and let M satisfy $|q_i| \leq sM$ for every quantized weight q_i . Then a sufficient worst-case condition for unsigned 64-bit safety of the full encoding calculation is

$$4sMN \leq 2^{64} - 1.$$

Equivalently,

$$N \leq \left\lfloor \frac{2^{64} - 1}{4sM} \right\rfloor.$$

Under this condition, both the final encoded score e and the intermediate quantity $\text{withOffset} = \text{partialSum} + z_s$ in Eq. 6 fit in `uint64`.

Proof. Partition the quantized weights into positive, negative, and zero sets with cardinalities N_+ , N_- , and N_0 , so that $N_+ + N_- + N_0 = N$. Because $g_i \in \{0, 1, 2\}$

and $z_w = \max(0, -\min_i q_i) \leq sM$, the shifted weight $u_i = q_i + z_w$ obeys

$$u_i \leq \begin{cases} 2sM, & q_i > 0, \\ sM, & q_i \leq 0. \end{cases}$$

Hence the encrypted accumulation before correction satisfies

$$\text{partialSum} = \sum_i g_i u_i \leq 4sMN_+ + 2sM(N_- + N_0).$$

The score zero-point satisfies

$$z_s = - \sum_{i: q_i < 0} 2q_i \leq 2sMN_-.$$

Therefore the intermediate value in Eq. 6 is at most

$$\begin{aligned} \text{withOffset} &= \text{partialSum} + z_s \\ &\leq 4sMN_+ + 4sMN_- + 2sMN_0 \\ &\leq 4sMN. \end{aligned}$$

This is the worst-case upper bound for the intermediate sum.

For the final encoded score, observe that the weight-shift correction exactly cancels the zero-point contribution:

$$P = \text{partialSum} - z_w G = \sum_i g_i q_i.$$

Thus

$$-2sMN_- \leq P \leq 2sMN_+,$$

and after adding z_s we obtain

$$0 \leq e = P + z_s \leq 2sM(N_+ + N_-) \leq 2sMN.$$

So the encoded score range itself fits in `uint64` under the weaker bound $2sMN \leq 2^{64} - 1$, but the actual implementation must also accommodate the larger intermediate `withOffset`. A sufficient safety condition for all intermediate and final values is

$$4sMN \leq 2^{64} - 1.$$

This condition was satisfied by every model evaluated in this study. It is an arithmetic safety check, not evidence that the system can process models larger than the evaluated maximum of 5,000 variants.

4.5 Selecting the scale factor

Before publishing a model, the quantization procedure evaluates the candidate scales

$$s \in \{10^2, 10^4, 10^6, 10^8, 10^{10}\}$$

against the actual GWAS weight distribution. For each s , it computes the mean absolute error (MAE) between the quantized dot product and the floating-point PRS for all individuals in the study data set. It compares three ranges:

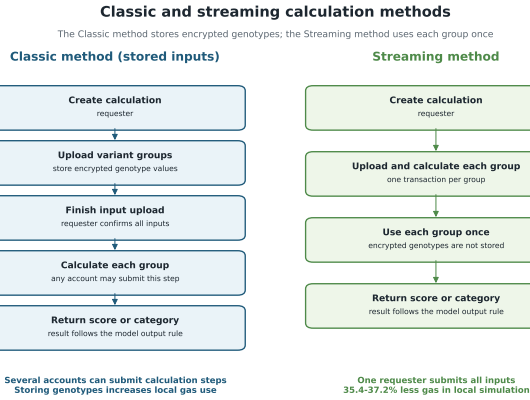


Figure 5: Calculation method comparison. The Classic method (stored inputs) uploads and stores SNPs before the score is calculated, so different parties can perform these steps. The Streaming method combines upload and calculation for one requester. Across the 100–5,000 variants evaluated in the local simulation, streaming used 35.4–37.2% less gas.

- **Low scale** ($s \approx 10^2$): 1–15% MAE in these data.
- **Selected scale** ($s \approx 10^6$): zero error on these six-decimal-place data sets. A candidate recommendation for similarly represented models, not a universal production default.
- **Higher scale** ($s \approx 10^8$): no improvement over the selected scale on the study data; the limiting factor is source data precision.

This calculation takes approximately 200 ms and should be performed before every model publication. The scale does not change gas use; the number of transactions needed to submit the SNPs is the limiting factor.

5 Calculation methods

bioETH-PRS provides two methods that differ in whether encrypted SNPs are stored between transactions (Figure 5). Both methods produce the same score.

5.1 Classic method (stored inputs)

The requester uploads encrypted SNPs in groups of at most 32. After upload, the score is calculated in groups of 20 SNPs, just below the largest successful local group of 21. The encrypted SNP references remain stored between transactions. This allows a different account, such as a relay, to continue the calculation. Under the encryption assumptions the relay cannot read the SNP values, but it can delay requests. Its transactions also reveal timing and addresses that are already visible on the blockchain. It therefore remains a dependency for availability and privacy.

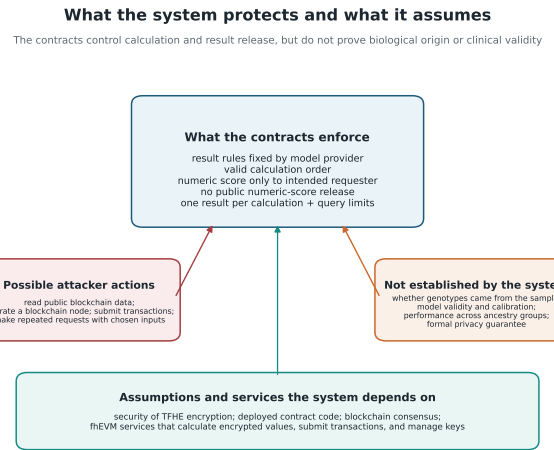


Figure 6: Security assumptions and remaining limits. The contracts control who may request a calculation, how the result is released, and how often a sample may be queried. They do not prove that the encrypted SNPs came from the registered sample, that the model is clinically valid, or that repeated queries reveal no information about private weights.

5.2 Streaming method

The requester provides and processes each group of up to 20 encrypted SNPs in the same transaction. The values are multiplied by the model weights, added to the running sum, and then discarded rather than stored for later transactions.

Gas savings. The Classic method permanently stores an encrypted SNP reference and a record stating who may use it. The Streaming method avoids both records, which explains the observed 35.4–37.2% reduction in local gas. The remaining gas in this implementation comes mainly from preparing each encrypted input and multiplying and adding encrypted values.

Trade-off. The Classic method allows upload and calculation to be performed by different parties. The Streaming method requires one requester to submit and process the inputs.

6 Security assumptions and limits

6.1 Attacker considered

We assume an attacker can read public blockchain data, submit transactions, and make repeated classification requests with chosen encrypted inputs (Figure 6). A requester who is allowed to use private weights may submit values that do not come from the registered sample. We assume that the attacker cannot break TFHE encryption or bypass the contract rules about who may use data and receive results.

6.2 Source of the encrypted SNPs

The registry checks whether a person may request a calculation, but it cannot check whether the encrypted SNP values came from the registered biological sample. The contracts calculate a result from the values they receive, including values chosen by a requester. Restricting an attack to correlated SNP patterns is therefore an analysis choice, not a restriction enforced by the system.

This study assumes that the patient, laboratory, or data holder prepares the genotype data correctly before encryption, following Section 2.2. The stored sample record describes the genome build, variant order, preparation rules, and a cryptographic fingerprint of that record. It helps document how the input was prepared, but it does not prove that the encrypted values came from the stated biological sample.

6.3 Rules for protecting results

Under the assumptions in Table 2, the contracts apply five rules to protect the results:

- Contracts store opaque references to encrypted values, not plaintext SNPs. The requester creates the encrypted inputs, and the fhEVM services process them.
- Each encrypted output records who may decrypt it, and the contract gives this record only after the required calculation steps.
- The encoded PRS e (Eq. 6) is never publicly decryptable.
- A model may require the randomized Low, Medium, or High category instead of releasing the raw score to the requester.
- Each calculation releases one result; another result requires another query.

These rules do not establish where the submitted SNPs came from, whether the model is clinically valid, or whether the external fhEVM services behave correctly.

6.4 Randomized risk category

Before assigning Low, Medium, or High risk, the Result contract adds a random nonnegative integer smaller than B to the encoded score:

$$e_{\text{randomized}} = e + \nu, \quad \nu \sim \text{DiscreteUniform}\{0, 1, \dots, B - 1\}. \quad (8)$$

The fhEVM supplies ν ; the requester does not choose it. The model provider sets B and the two category thresholds before any request is made, and requesters cannot change the thresholds between queries. This makes it harder to narrow down an encrypted score by repeatedly changing a comparison threshold.

All comparisons against classification thresholds operate on the randomized encrypted score using encrypted

boolean operations, avoiding any branching on hidden values:

$$\begin{aligned} \hat{L} &= \mathbb{K}[e_{\text{randomized}} < \tau_L], \\ \hat{H} &= \mathbb{K}[e_{\text{randomized}} \geq \tau_H], \\ \text{cat} &= \text{select}(\hat{L}, L, \text{select}(\neg \hat{L} \wedge \neg \hat{H}, M, H)). \end{aligned} \quad (9)$$

The comparisons are carried out on encrypted values. The two thresholds must be at least B apart.

Interpretation and limits. This method does not provide an (ϵ, δ) -differential privacy guarantee. It uses one-sided noise, does not relate B to a formal measure of how much one input can change the score, and does not account for information from repeated queries. A formal privacy method would need to address each of these points.

Bias correction. Because the random value is always nonnegative, its exact mean is $(B - 1)/2$. The contract provides the integer threshold correction $B/2$; for $B = 128$, this is 64, while the exact mean is 63.5. Scores close to a threshold remain uncertain. In the 100-SNP study, 48 individuals outside this uncertainty range kept the same category. The other two were each 64 units below a threshold, so their category could change.

6.5 Analysis of repeated queries

We examined whether a requester allowed to use a model with 20 private weights could infer the weights by submitting chosen encrypted SNP values and observing the released results. The analysis included thresholds chosen by the requester, with each query chosen after seeing earlier results or all queries chosen in advance; thresholds fixed by the model provider; several wallets and samples; and correlated SNP patterns. These calculations were performed in the local simulation. To make the analysis repeatable, we used a fixed sequence of random integers from 0 through 127. A different random sequence may give different exact recovery counts. The contract also limits the number of requests for each model, requester, and sample over a set number of blocks.

In this analysis, changing the threshold after each result recovered 19 of 20 weights within B after 200 queries and first recovered all 20 after 260 queries. When all queries were chosen in advance, none was recovered within B after 320 queries ($r = 0.6689$; 65% of signs correct). With thresholds fixed by the model provider, none of the 20 estimates was within B after 320 queries. However, Pearson $r = 0.9388$ and 70% sign accuracy show that the results still revealed some information about the relative weights. The fixed thresholds therefore made precise recovery harder but did not completely hide the private weights.

Using several wallets did not increase the number of queries allowed for the same sample, although different registered samples had separate limits. For private weights, the model provider decides who may use them. When inputs were restricted to correlated SNP blocks,

Table 2: Parts of the system on which privacy, correctness, availability, and the sample or model record depend. A mark shows what could be affected if that part fails.

Component	Privacy	Correctness	Availability	Record	Responsibility
Genotype provider/preprocessor		X		X	prepares the claimed sample and applies build, order, QC, and orientation rules
Model provider		X		X	publishes the intended weights, metadata, thresholds, and scientifically valid model
Smart contracts	X	X	X	X	apply the stated calculation and the rules about who may use data or receive results
Blockchain consensus		X	X	X	orders and finalizes the publicly auditable contract record
fhEVM coprocessor	X	X	X		performs TFHE operations correctly without exposing plaintext
Relayer	X		X		carries requests without revealing a released result to someone else
Decryption services	X	X	X		decrypt results only for the intended requester

Table 3: Results of the repeated-query analysis in the local simulation. “Within B ” is the number of weights estimated to within the randomization range.

Information released and query choice	Queries	Pearson r	Sign acc.	Within B
Raw score available	20	1.0000	100%	20/20
Requester changes threshold after each result	200	0.9999	100%	19/20
Requester changes threshold after each result, first 20/20	260	1.0000	100%	20/20
Requester-selected thresholds, queries chosen in advance	320	0.6689	65%	0/20
Thresholds fixed by model provider	320	0.9388	70%	0/20
Fixed thresholds with correlated SNP blocks	320	0.0223	65%	0/20

the correlation between estimated and true weights fell to 0.0223. This is not a reliable safeguard because requesters can still submit other encrypted SNP values. Applying the studied limit of three calculations per 1,000 blocks to 260 total queries gives calculated times of 288.9 hours with 12-second blocks or 48.1 hours with 2-second blocks for one registered sample. These are calculated examples, not measured network times. The chosen $B = 128$ was 1.34% of the largest integer weight; other models may require a different value.

7 Evaluation

7.1 Where calculations were evaluated

We report results from Sepolia and from a local contract simulation. Sepolia provides observed fhEVM transaction results. The local simulation evaluates contract behavior, transaction counts, and gas used in a local contract environment; it does not measure real encrypted-computation time, Sepolia capacity, or production fees.

We evaluated public-weight local calculations with 100, 500, 1,000, and 5,000 SNPs [Knight et al., 2026], and one private-weight local calculation with 100 SNPs. Each public data set also includes one leading constant position. The largest successful calculation chunk in the local sim-

ulation contained 21 SNPs for both public and private weights, and we used a chunk size of 20. The corresponding Sepolia limit was not measured.

On Sepolia (chain ID 11155111), a public-weight 100-SNP calculation completed in 25 transactions after four contract deployments. It used 20.710271 million gas, took 269,320 ms from submission to result and 8,081 ms to decrypt, and returned an encoded score of 758,685. This score exactly matched the independent calculation. The private-weight 100-SNP calculation was evaluated locally but was not run on Sepolia.

7.2 Agreement with an independent calculation

We compared all 200 public-weight results with an independent calculation of Equation 1: 50 individuals at each of the four model sizes. The mean, root-mean-square, and maximum absolute errors were all zero; all 200 scores matched exactly; and Pearson $r = 1$. This supports the calculation for the studied inputs.

The exact agreement is not a general precision result. The four data sets contain 6,600 variant weights plus four leading constant weights. All 6,604 values have at most six decimal places, and the selected scale represented them without rounding. Other weight sets may show small rounding differences.

Table 4: Sepolia and local results for the same public-weight 100-SNP calculation using the Classic method (stored inputs), upload groups of 32, calculation groups of 10, and 25 transactions. Sepolia used 10.42% more gas than this one local simulation; this percentage is not a general conversion between local and network results.

Setting	Tx	Gas	Submit-result	Decrypt	Encoded score
Sepolia	25	20.710271 M	269,320 ms	8,081 ms	758,685
Local simulation	25	18.755864 M	147 ms	125 ms	758,685

Table 5: Independent Equation 1 calculation versus decoded public-weight bioETH-PRS in the local simulation. The study weights were represented without rounding.

SNPs	Scale s	n	MAE/RMSE/max	Exact
100	3×10^6	50	0/0/0	50/50
500	3×10^6	50	0/0/0	50/50
1,000	1×10^6	50	0/0/0	50/50
5,000	1×10^6	50	0/0/0	50/50

Table 6: Transactions for the public-weight calculation on Sepolia and in the local simulation.

Setting	Variants	Method	Tx
Sepolia	100	Classic	25
Local simulation	100	Streaming	15
Local simulation	500	Streaming	47
Local simulation	1,000	Streaming	88
Local simulation	5,000	Streaming	413

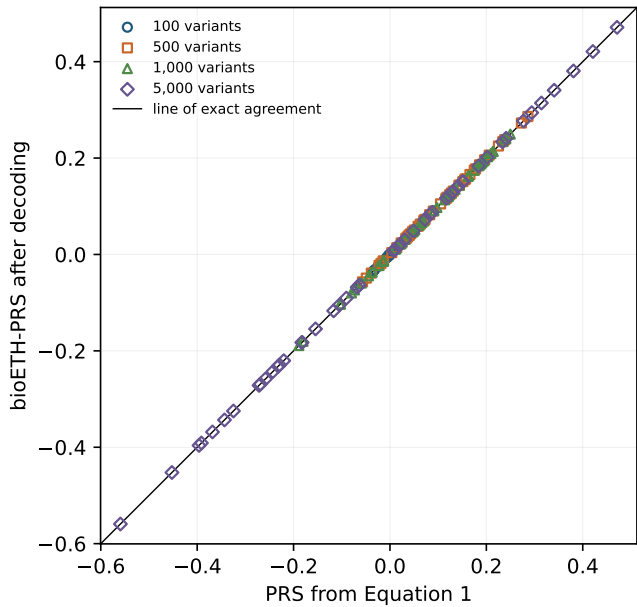


Figure 7: Equation 1 PRS versus decoded public-weight bioETH-PRS for all 200 individual-model pairs in the local simulation. Every point lies on the line of exact agreement; overplotting is reduced by model-specific markers.

7.4 Gas use as variant count increases

Figure 8 and Table 7 report gas used by the two calculation methods for four public-weight calculations in the local simulation. Gas grows approximately linearly over the measured 100–5,000-variant range; this is not a claim about scaling on Sepolia. Across this range, the Streaming method used 35.4–37.2% less gas because it did not store each encrypted SNP reference permanently.

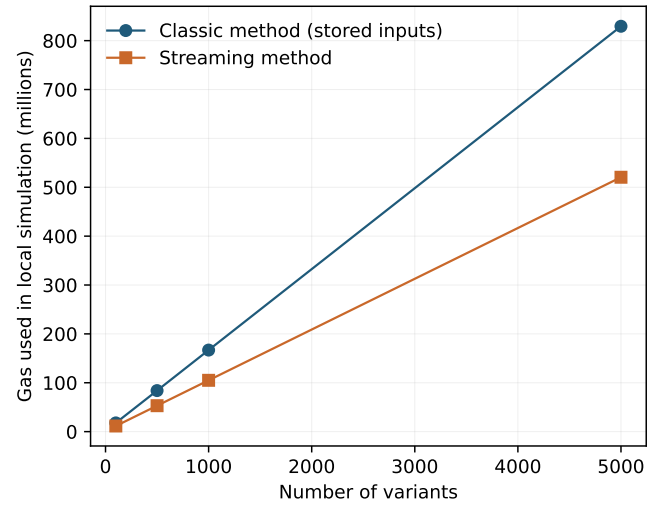


Figure 8: Gas used versus variant count for both calculation methods in the local simulation. The Streaming method used 35.4–37.2% less gas because it did not store each encrypted SNP handle permanently.

For the randomized risk category at 100 SNPs, all 48 individuals outside the threshold uncertainty range kept the same category. Two individuals were inside the range and are reported separately; one changed category in this calculation.

7.3 Variant scale

Table 6 reports the number of transactions used for one public model and one sample. The 100-SNP Sepolia result used the Classic method (stored inputs). The local results from 100 to 5,000 variants used the Streaming method; 5,000 variants was the largest bioETH-PRS model evaluated.

7.5 Gas used for each SNP

Table 8 breaks down the approximate gas used for each SNP in the local simulation. In the present implementation, preparing the encrypted input and multiplying and adding encrypted values account for approximately 95,000–104,000 gas per SNP before permanent storage

Table 7: Gas used in one local calculation, rounded to 0.001 million. The classic and streaming totals include model publication and result release but exclude sample registration. Both methods used calculation chunks of 20 SNPs.

SNPs	Classic	Streaming	Reduction
100	17.863 M	11.538 M	35.4%
500	83.985 M	53.063 M	36.8%
1,000	166.875 M	105.013 M	37.1%
5,000	829.422 M	520.487 M	37.2%

is considered. Changes to the fhEVM services or input preparation could change these values.

Table 8: Approximate gas used for each SNP in the local simulation

Operation	Classic	Streaming
Prepare encrypted input	~50,000	~50,000
Multiply encrypted values	~27,000	~27,000
Add encrypted values	~27,000	~27,000
Store encrypted SNP (classic only)	~25,000	—
Store who may use it (classic only)	~25,000	—
Read stored data and other overhead	~12,000	—
Total/SNP	~166,000	~104,000

7.6 Latency

Local-simulation times for each public-weight calculation were 157, 780, 1,672, and 8,819 ms for 100, 500, 1,000, and 5,000 variants, respectively. Each data set also includes one leading constant position. The measured range was 1.554–1.763 ms per position. The time per position increased by about 13% between the smallest and largest local calculations. These times combine the local arithmetic and transaction overhead; they measure neither real TFHE evaluation nor network latency. The separate Sepolia 100-SNP times appear in Table 4 and are not compared as a speed ratio.

7.7 Transactions, gas, and fee examples

Table 9 reports the observed transaction use. The Sepolia deployment used four transactions and 5.892559 million gas, costing 0.0062781714 Sepolia test ETH; the public-weight calculation used 25 transactions and 20.710271 million gas, costing 0.0252747648 test ETH. These are test-network expenditures, not production prices. In the local simulation, a public-weight calculation with 100 SNPs using the Streaming method used 15 transactions and 11.690 million gas, whereas the private-weight calculation with 100 SNPs used 17 transactions and 23.508 million gas (2.01×). The private-weight calculation is the one relevant to attempts to recover the weights. Decryption happens outside Ethereum and requires no additional Ethereum transaction.

Table 9: Measured transaction use. The local streaming totals include sample registration and come from a separate calculation from Table 7; encrypted inputs also cause small gas variation between otherwise identical calculations.

Setting	Quantity	Tx	Gas
Sepolia	four-contract deployment	4	5.892559 M
Sepolia	public-weight 100-SNP, Classic method	25	20.710271 M
Local simulation	public-weight Streaming method	100-SNP, 15	11.690 M
Local simulation	private-weight Streaming method	100-SNP, 17	23.508 M
Local simulation	one-time category settings	1	0.077 M

Recording a digital fingerprint for the model file and another for its preparation information added 40,568 local gas per model. This added no encrypted arithmetic and did not change the largest successful local calculation group. The randomized category settings require one additional model transaction (77,314 local gas). Releasing a raw score used 169,898 local gas, whereas releasing a randomized category used 432,230 local gas. These are alternative ways to release one result and are not added together.

For illustration only, we multiplied three local gas measurements by hypothetical gas prices: deployment (5,892,613 gas), the public streaming calculation (11,690,021 gas), and the private streaming calculation (23,507,880 gas). These are calculated estimates, not observed costs. At 1 gwei, the respective values are 0.005892613, 0.011690021, and 0.02350788 ETH; at 30 gwei, they are 0.17677839, 0.35070063, and 0.7052364 ETH. We make no USD conversion or conclusion about affordability or clinical or commercial use.

7.8 Calculation checks and responsibilities

The independent comparison in Figure 7 shows agreement with Equation 1 for the studied inputs; it is not a guarantee for every model or deployment. Table 10 explains who is responsible for each part of the result.

The contracts check who may receive a result, allow only one result per calculation, keep the thresholds fixed by the model provider, require the minimum threshold gap, apply query limits, and stop a private-weight calculation when the model provider no longer allows it. They do not establish the source of the biological sample, clinical validity, calibration, or performance in other populations.

8 Who may use the data and calculation order

8.1 Encrypted values and released results

Each encrypted value has a contract record stating who may use or decrypt it. In the Classic method (stored inputs), the SNP handles and these records are stored for later transactions. In the Streaming method, they are

Table 10: Responsibilities and remaining limits.

Actor/component	Responsibility in this study	Remaining limit
Data preparer	check genome build, variant order, dosage validity, missing values, and effect-allele alignment	the contracts do not prove that the encrypted SNPs came from the stated biological sample
Model provider	supply the weights, model details, thresholds, and scientific justification	calibration or performance in populations that were not studied
Smart contracts	apply the stated integer calculation and rules about who may use data or receive results	the contracts cannot verify input claims or external services
fhEVM services	perform the encrypted arithmetic and release the result to the intended requester	blockchain consensus does not independently verify the encrypted arithmetic
Independent calculation	compare Equation 1 with 200 studied inputs	agreement is not a proof for other models, inputs, or deployments
Requester	inspect the published model details and blockchain transactions	clinical interpretation beyond the published model

kept only for the current transaction and then discarded. Avoiding these two permanent records for each SNP explains the lower gas use of the Streaming method. The encoded score is released only to the intended requester. If the model requires a risk category, only the randomized Low, Medium, or High result is made public.

8.2 Order of the calculation

The contracts require the operations to occur in order: create a calculation, provide the encrypted SNPs, calculate the weighted sum, and release one result. The Classic and Streaming methods cannot be mixed within the same calculation.

8.3 Who may use encrypted weights

For private weights, the model provider chooses which requesters may use the model. The contract checks this choice whenever a calculation group is processed. If the provider no longer allows a requester to use the model, the next group is rejected. The contract also checks who may use the registered sample when the calculation is created.

9 Discussion

We evaluated bioETH-PRS with additive PRS models containing up to 5,000 variants. Publicly auditable smart contracts reduce reliance on a single designated evaluator, but the system still depends on the blockchain and on fhEVM services that perform encrypted calculations, submit transactions, and decrypt results for the intended requester. A public-weight 100-SNP calculation completed on Sepolia. Public-weight calculations with 100–5,000 variants and one private-weight 100-SNP calculation were evaluated in the local simulation. These results do not establish genome-wide or clinical use.

9.1 Comparison with HEPRS

bioETH-PRS and HEPRS [Knight et al., 2026] address different questions and are best understood as complementary systems.

HEPRS reports encrypted PRS calculation for a substantially larger model and measures CKKS performance under a three-party non-collusion assumption. CKKS represents approximate signed decimal values without the integer conversion used here. HEPRS therefore provides the relevant results for large variant counts and measured FHE performance.

bioETH-PRS instead studies whether public smart contracts can coordinate the calculation and control result release without a designated evaluator. Its model settings and transaction history are public, while the system still relies on the fhEVM services. It also requires many blockchain transactions. The two systems answer different questions, and these results do not show that either is broadly superior.

9.2 Limitations

Variant scale. The largest successful calculation chunk in the local simulation contained 21 SNPs for both public and private weights; the Sepolia limit remains unknown. A public-weight 5,000-variant calculation required 413 local transactions. We did not evaluate larger bioETH-PRS models or production feasibility.

Source of the SNP data. As specified in Section 6, being allowed to request a calculation does not prove that the encrypted SNP values came from the registered sample.

URI observability. Sample data URIs are visible on the blockchain. Any network participant can read them, even if the application limits who may use the sample. Storing only a cryptographic fingerprint of the URI would reduce this exposure.

One-sided randomization and bias. Adding an integer chosen uniformly from 0 through $B - 1$ increases the score by $(B - 1)/2$ on average and creates an uncertainty

range near each threshold. This randomized category result is not differential privacy because the random value is one-sided, is not formally calibrated to score sensitivity, and is not analyzed across repeated queries.

Production cost. We report gas used by the contracts, Sepolia test-ETH expenditure, and calculated fee examples using hypothetical gas prices. Production fee schedules, USD cost, memory use, and operational throughput were not measured, so whether the system is affordable or practical for clinical or commercial use remains unknown.

9.3 Future work

Future work should reduce the number of blockchain transactions and evaluate the private-weight 100-SNP calculation on Sepolia. Establishing the source of encrypted SNPs will require a signed laboratory record or a privacy-preserving proof that links the encrypted values to the registered sample. A formal privacy guarantee will require a precise definition of which data sets are compared, how much one input can change the score, how noise is chosen, and what repeated queries reveal. Later studies could also evaluate LD-aware scores [Privé et al., 2021], decryption shared among several parties, model versioning, and reuse of encrypted SNP values across models.

10 Related Work

Privacy-preserving genomic computation has been pursued through multiple cryptographic paradigms. Kim and Lauter [Kim and Lauter, 2015] applied homomorphic encryption schemes to compute GWAS statistics for up to 5,000 sequences, but did not extend to PRS computation. Blatt et al. [Blatt et al., 2020] built a CKKS-based privacy-preserving GWAS calculation for large-scale studies, noting that PRS computation from decrypted summary statistics was future work. McLaren et al. [McLaren et al., 2016] demonstrated privacy-preserving genomic testing in an HIV clinical-care model using homomorphic encryption, but did not address the full PRS calculation. Raisaro et al. [Raisaro et al., 2018] introduced MedCo for secure, privacy-preserving exploration of distributed clinical and genomic data.

Knight et al. [Knight et al., 2026] (HEPRS) is the closest prior work, providing the first complete FHE calculation of PRS with real clinical data. HEPRS achieves $r > 0.999$ correlation with plaintext scores on a 110,000-SNP schizophrenia model and demonstrates practical feasibility on commodity hardware. bioETH-PRS builds directly on the HEPRS computational framework and uses the same GWAS study data, studying a different trade-off in which public contracts coordinate the calculation while infrastructure trust remains.

On the blockchain side, prior work on privacy-preserving smart contracts has focused on anonymous payments and related protocol work [Ben-Sasson et al., 2014, Pertsev

et al., 2019] and zero-knowledge proof-based computation [Ben-Sasson et al., 2018]. Zero-knowledge proofs can verify a calculation while keeping designated witness data private, whereas FHE allows calculations directly on encrypted data. FHE on the blockchain, typified by the fhEVM framework [Zama, 2025], combines confidential operations with public smart-contract records while depending on external encrypted-computation services. bioETH-PRS applies this framework to polygenic risk score calculation.

11 Conclusion

We presented bioETH-PRS, a protocol for confidential polygenic risk scoring using TFHE-based fully homomorphic encryption on a programmable blockchain. Publicly auditable smart contracts coordinate the calculation without a designated evaluator. This reduces reliance on one evaluator but does not remove trust: the result still depends on genotype preparation, model validity, the contracts, blockchain, and fhEVM services that perform encrypted calculations and decrypt results for the intended requester.

One public-weight 100-SNP calculation completed on Sepolia and matched the independent calculation exactly. Public-weight calculations with 100–5,000 variants and one private-weight 100-SNP calculation were evaluated only in the local simulation. All 200 public-weight local scores matched Equation 1 exactly because the study weights were represented without rounding. The Streaming method used 35.4–37.2% less local gas. In the repeated-query analysis, thresholds fixed by the model provider and query limits made precise weight recovery harder, but some information about relative weights remained. The randomized category result does not provide differential privacy.

The results apply to additive PRS models containing up to 5,000 variants. This study does not establish genome-wide or clinical use, production affordability, private-weight calculation on Sepolia, or the largest calculation step that Sepolia can support. HEPRS demonstrates encrypted calculation at a larger scale; bioETH-PRS studies how smart contracts can coordinate a smaller calculation and control result release.

Biographical Note

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Data Availability Statement

No new datasets were generated or analyzed in this study.

Code Availability Statement

All relevant code and results can be found on GitHub at <https://github.com/Georgakopoulos-Soares-lab/bioETH-PRS>.

References

- E. Ben-Sasson, A. Chiesa, C. Garman, M. Green, I. Miers, E. Tromer, and M. Virza. Zerocash: Decentralized anonymous payments from bitcoin. In *2014 IEEE Symposium on Security and Privacy*, pages 459–474, 2014. doi: 10.1109/SP.2014.36.
- E. Ben-Sasson, I. Bentov, Y. Horesh, and M. Riabzev. Scalable, transparent, and post-quantum secure computational integrity. Cryptology ePrint Archive, Report 2018/046, 2018. URL <https://eprint.iacr.org/2018/046>.
- M. Blatt, A. Gusev, Y. Polyakov, and S. Goldwasser. Secure large-scale genome-wide association studies using homomorphic encryption. *Proceedings of the National Academy of Sciences*, 117(21):11608–11613, 2020. doi: 10.1073/pnas.1918257117.
- C. C. Chang, C. C. Chow, L. C. Tellier, S. Vattikuti, S. M. Purcell, and J. J. Lee. Second-generation PLINK: Rising to the challenge of larger and richer datasets. *GigaScience*, 4:7, 2015. doi: 10.1186/s13742-015-0047-8.
- J. H. Cheon, A. Kim, M. Kim, and Y. Song. Homomorphic encryption for arithmetic of approximate numbers. In *Advances in Cryptology – ASIACRYPT 2017*, volume 10624 of *Lecture Notes in Computer Science*, pages 409–437, 2017. doi: 10.1007/978-3-319-70694-8_15.
- I. Chillotti, N. Gama, M. Georgieva, and M. Izabachene. TFHE: Fast fully homomorphic encryption over the torus. *Journal of Cryptology*, 33:34–91, 2020. doi: 10.1007/s00145-019-09319-x.
- Y. Erlich and A. Narayanan. Routes for breaching and protecting genetic privacy. *Nature Reviews Genetics*, 15:409–421, 2014. doi: 10.1038/nrg3723.
- M. Fredrikson, S. Jha, and T. Ristenpart. Model inversion attacks that exploit confidence information and basic countermeasures. In *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security*, pages 1322–1333, 2015. doi: 10.1145/2810103.2813677.
- M. Gymrek, A. L. McGuire, D. Golan, E. Halperin, and Y. Erlich. Identifying personal genomes by surname inference. *Science*, 339(6117):321–324, 2013. doi: 10.1126/science.1229566.
- M. Inouye, G. Abraham, C. P. Nelson, et al. Genomic risk prediction of coronary artery disease in 480,000 adults. *Journal of the American College of Cardiology*, 72(16):1883–1893, 2018. doi: 10.1016/j.jacc.2018.07.079.

- M. Kim and K. Lauter. Private genome analysis through homomorphic encryption. In *Financial Cryptography and Data Security*, pages 239–256, 2015. doi: 10.1007/978-3-662-48185-7_14.
- E. Knight, J. Li, M. Jensen, I. Yolou, C. Kockan, and M. Gerstein. Homomorphic encryption enables privacy preserving polygenic risk scores. *Cell Reports Methods*, 6:101271, 2026. doi: 10.1016/j.crmeth.2025.101271.
- S. A. Lambert, G. Abraham, and M. Inouye. Towards clinical utility of polygenic risk scores. *Human Molecular Genetics*, 28(R2):R133–R142, 2019. doi: 10.1093/hmg/ddz187.
- P. J. McLaren, J. L. Raisaro, M. Aouri, et al. Privacy-preserving genomic analysis for personalized HIV treatment. In *Proceedings of the 7th International Workshop on Data Privacy in Healthcare*, pages 1–10, 2016. doi: 10.1145/2983655.2983659.
- A. Pertsev, R. Semenov, and R. Storm. Tornado cash privacy solution. Protocol white paper, 2019. URL https://tornado.cash/audits/TornadoCash_whitepaper_v1.4.pdf.
- F. Privé, J. Arbel, and B. J. Vilhjálmsón. LDpred2: Better, faster, stronger. *Bioinformatics*, 36(22–23):5424–5431, 2021. doi: 10.1093/bioinformatics/btaa1029.
- J. L. Raisaro, G. Choi, S. Pradervand, et al. Protecting privacy and security of genomic data in i2b2 with homomorphic encryption and differential privacy. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 15(5):1413–1426, 2018. doi: 10.1109/TCBB.2018.2854782.
- Tune Insight. Lattigo v5. Open-source Go library, 2023. URL <https://github.com/tuneinsight/lattigo>.
- N. R. Wray, T. Lin, J. Austin, J. J. McGrath, I. B. Hickie, G. K. Murray, and P. M. Visscher. From basic science to clinical application of polygenic risk scores: A primer. *JAMA Psychiatry*, 78(1):101–109, 2021. doi: 10.1001/jamapsychiatry.2020.3049.
- Zama. fhEVM: Confidential smart contracts using fully homomorphic encryption. Official documentation, 2025. URL <https://docs.zama.ai/fhevm>.